

Short Spec — MRTP Convexity Layer

Short vs Put — Recoverable vs Irreversible Convexity

1. Core Principle

PolyAgora currently governs:

- admissibility,
- recoverability,
- drawdown compression,
- crisis maneuverability.

Phase II adds:

exposure recoverability governance.

The system must distinguish:

Type	Meaning
Irreversible Convexity	convexity that damages future maneuverability
Recoverable Convexity	convexity that preserves optionality and future steering

2. Short vs Put — MRTP Interpretation

SHORT

Shorting assumes:

“future re-entry into ownership remains admissible.”

Meaning:

- future liquidity must exist,
- future buyback corridor must remain open,
- squeeze risk must remain bounded.

Thus shorts are:

- forced-traversal dependent,
- corridor fragile,
- potentially non-recoverable.

M RTP Classification

Usually:

$$SHORT \rightarrow Low ERQ$$

Especially during:

- fragmentation,
- transition instability,
- squeeze regimes,
- liquidity rupture.

PUT

Put structures:

- bound downside,
- preserve optionality,
- do not require forced future traversal,
- maintain maneuverability under rupture.

M RTP Classification

Usually:

$$PUT \rightarrow High ERQ$$

Especially during:

- transition,
- ridge pressure,
- corridor narrowing,
- volatility expansion.

3. New Runtime Variable

Introduce:

$$ERQ_i \in [0, 1]$$

Exposure Recoverability Quality.

Measures:

- forced traversal dependency,

- liquidity survivability,
 - boundedness,
 - optionality preservation,
 - corridor robustness.
-

4. Runtime Rule

Current PolyAgora:

- governs regime admissibility.

Phase II:

- governs exposure admissibility.

Meaning:

$$Allocation = f(Zone, SM, Corridor, ERQ)$$

Not merely:

- bullish/bearish.

But:

- recoverable vs irreversible exposure.
-

5. Two Convexity Classes

A. Irreversible Convexity

Examples:

- naked leverage,
- forced shorts,
- concentrated beta,
- liquidity-fragile momentum.

Properties:

- future forced traversal required,
- collapses under corridor rupture.

Policy:

- suppress during transition/boundary regimes.
-

B. Recoverable Convexity

Examples:

- long puts,
- bounded asymmetry,
- optionality sleeves,
- dispersion,
- recoverable trend structures.

Properties:

- preserve maneuverability,
- bounded downside,
- maintain re-entry capacity.

Policy:

- preferred during transition/expansion.

6. Historical Backtest Role

What we identify post-record

We evaluate:

Metric	Meaning
Corridor survivability	did exposure preserve maneuverability?
Re-entry preservation	could system rotate post-rupture?
Forced deleveraging risk	was future traversal destroyed?
Convex participation	did system capture admissible upside?
Recovery latency	how fast could system re-expand?

Historical classification

Example:

Momentum during COVID rebound

High convexity,

BUT:

- partially irreversible,
- high crash sensitivity.

Long-vol / put structures during COVID

Lower carry,

BUT:

- highly recoverable,
 - preserved steering.
-

7. Real-Time System Integration

This is the important part.

The edge is NOT:

- post-facto labeling.

The edge is:

real-time corridor geometry detection.

8. Real-Time Runtime Logic

LOCAL STAR / STABLE EXPANSION

If:

- fragmentation low,
- corridor width widening,
- ridge pressure falling,

then:

- permit recoverable convexity expansion,
- increase Kelly gradually.

Examples:

- controlled momentum,
 - dispersion,
 - optionality-preserving convex sleeves.
-

TRANSITION REGIME

If:

- corridor narrowing,

- fragmentation rising,
- ridge pressure increasing,

then:

- suppress irreversible convexity,
- rotate toward recoverable convexity.

Meaning:

- avoid leverage,
- avoid forced shorts,
- favor puts / bounded asymmetry.

BOUNDARY / RUPTURE

If:

- recoverability collapsing,

then:

- disable irreversible exposure,
- preserve optionality only,
- maintain future maneuverability.

9. Real-Time Signals

The system monitors:

Signal	Meaning
Corridor width	future maneuverability
Ridge pressure	transition instability
Fragmentation	transition-law deformation
Liquidity deformation	forced traversal danger
Correlation rupture	regime instability
Survival matrix decay	admissibility collapse

These now govern:

- WHICH convexity is allowed.

Not merely:

- risk-on / risk-off.

10. Kelly Integration

Kelly no longer expands purely on expected return.

Instead:

$$Kelly_t = f(Return, Recoverability, ERQ, CorridorWidth)$$

Meaning:

- leverage allowed only when future maneuverability survives.
-

11. Core MRTP Upgrade

Old PolyAgora:

survive crises.

New PolyAgora:

expand only through trajectories that preserve future steering.

This is the bridge between:

- MRTP,
 - short vs put,
 - convexity governance,
 - real-time regime steering.
-

12. Final CTO Objective

Goal is NOT:

- maximum beta,
- maximum momentum,
- maximum CAGR.

Goal is:

preserve the crisis moat while selectively admitting recoverable convexity during admissible expansion regimes.

This becomes:

- the next architecture layer,
- the explanation of current V10 behavior,
- and the roadmap toward Phase II.